

CFD Assignment 4

Ambuj Nayan

24M0003

Aerospace Engineering

Indian Institute of Technology Bombay

Problem Statement We consider the steady flow of a calorically perfect gas with specific heat ratio $\gamma = 1.4$ through a converging-diverging (CD) nozzle. The nozzle geometry is defined over the domain $0 \leq x \leq 2$, with a prescribed cross-sectional area distribution given by:

$$A(x) = 1.0 + 2.0(x - 1)^2 \quad (1)$$

The inlet (reservoir) conditions are a total (stagnation) pressure of $p_0 = 1.0133 \times 10^5$ Pa and a total temperature of $T_0 = 300$ K. The flow is driven by specifying an exit-to-stagnation pressure ratio of $p_e/p_0 = 0.585$.

To numerically solve the governing equations, we apply the **van Leer Flux Vector Splitting (FVS)** method to the quasi-one-dimensional Euler equations. The numerical solution captures both the expansion and the non-isentropic recompression due to shock.

The objectives of the present study are as follows:

- Solve the governing quasi-1D Euler equations using van Leer's FVS method for the specified area distribution and boundary conditions.
- Plot the non-dimensional pressure p/p_0 and Mach number as functions of axial distance x .
- Compute the exact quasi-1D results and compare with obtained numerical results.

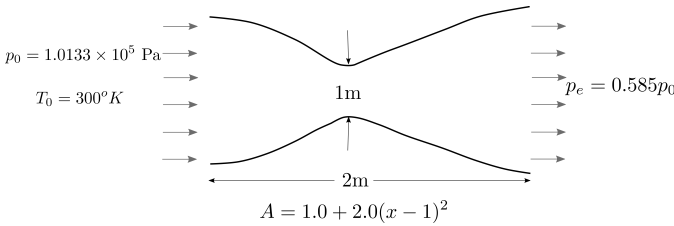


Fig. 1: Problem Statement

A. Governing Equations

For quasi-one-dimensional (quasi-1D) nozzle flows, the governing Euler equations describe the conservation of mass, momentum, and energy while accounting for variations in cross-sectional area. These equations are derived from the full three-dimensional Euler equations under the assumptions of axial symmetry and small flow angle variations. The conservative form is given as:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}$$

where the conservative state vector $\mathbf{U}$, the flux vector $\mathbf{F}$, and the source term vector $\mathbf{S}$ are defined as:

$$\mathbf{U} = \begin{bmatrix} \rho A \\ \rho u A \\ EA \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u A \\ (\rho u^2 + p)A \\ u(E + p)A \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ p \frac{dA}{dx} \\ 0 \end{bmatrix}$$

B. Numerical Scheme

The van Leer Flux Vector Splitting (FVS) method is used to solve the quasi-1D Euler equations numerically.

C. Discretized Governing Equation

Applying a first-order upwind scheme to the conservative form gives the time-marching discretized equation:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x} (\mathbf{F}_i^{+n} - \mathbf{F}_{i-1}^{+n}) - \frac{\Delta t}{\Delta x} (\mathbf{F}_{i+1}^{-n} - \mathbf{F}_i^{-n}) + \Delta t \mathbf{S}_i$$

The source term is discretized as:

$$\mathbf{S}_i = \begin{bmatrix} 0 \\ p_i \frac{A_{i+1} - A_{i-1}}{2\Delta x} \\ 0 \end{bmatrix}$$

D. Flux Splitting for van Leer Method

Split fluxes depend on the local Mach number M and are defined as follows:

- For $M \leq -1$:

$$\mathbf{F}^+ = 0, \quad \mathbf{F}^- = \mathbf{F}$$

- For $-1 < M < 1$:

$$\mathbf{F}^+ = \frac{1}{4} \rho a (M+1)^2 A \begin{bmatrix} 1 \\ \frac{2a}{\gamma} (1 + \frac{\gamma-1}{2} M) \\ \frac{2a^2}{\gamma^2-1} (1 + \frac{\gamma-1}{2} M)^2 \end{bmatrix}, \quad \mathbf{F}^- = \mathbf{F} - \mathbf{F}^+$$

- For $M \geq 1$:

$$\mathbf{F}^+ = \mathbf{F}, \quad \mathbf{F}^- = 0$$

where $a = \sqrt{\gamma RT}$ is the local speed of sound, and $\gamma = 1.4$ for air.

E. Time Step from Stability Condition

The time step Δt is computed using a CFL-based stability condition:

$$\Delta t = \nu \frac{\Delta x}{\lambda_{\max}}, \quad \lambda_{\max} = \max_i (|u_i| + a_i)$$

where $\nu \leq 1$ is the Courant number, typically less than or equal to 1.

F. Grids and Initialization

Use $I_{\max} = 101$ grid points along the nozzle. Initialize the flow field based on inlet stagnation conditions.

G. Boundary Conditions

Inlet: The stagnation pressure and temperature are specified. The velocity is extrapolated from the interior: $u_1 = u_2$. Pressure, density, and temperature are determined using isentropic relations.

Outlet: The exit pressure is fixed as $p_e/p_0 = 0.585$. Velocity and density are extrapolated from the interior:

$$u_{I_{\max}} = u_{I_{\max}-1}, \quad \rho_{I_{\max}} = \rho_{I_{\max}-1}$$

I. IMPLEMENTATION

A. Analytical Approach

The exit Mach number M_e can be determined using the following equation [1, Equation-5.28]:

$$M_e^2 = -\frac{1}{\gamma-1} + \sqrt{\frac{1}{(\gamma-1)^2} + \left(\frac{2}{\gamma-1}\right) \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left(\frac{p_{oe} A_e^*}{p_e A_e}\right)^2} \quad (2)$$

The term $\frac{p_{oe} A_e^*}{p_e A_e}$ can be expressed as [1, Equation-5.24]:

$$\frac{p_e A_e}{p_{oe} A_e^*} = \frac{p_e A_e}{p_{oe} A_t^*} = \left(\frac{p_e}{p_{oe}}\right) \left(\frac{A_e}{A_t^*}\right) \quad (3)$$

In Eq. (3), $\frac{p_e}{p_{oe}}$ represents the isentropic pressure ratio across the nozzle, while $\frac{A_e}{A_t^*}$ is the known exit-to-throat area ratio. As both terms are known, the quantity $\frac{p_e A_e}{p_{oe} A_e^*}$ is determined.

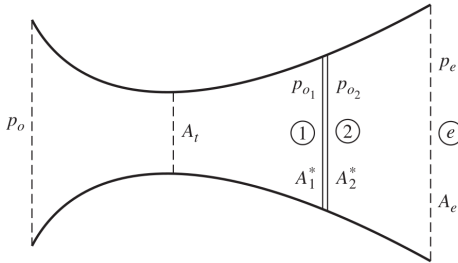


Fig. 2: Schematic of a normal shock inside a nozzle with station markings.

The procedure to determine the location of the normal shock inside the nozzle is outlined below:

- 1) Putting M_e into Eq. 8, the isentropic pressure ratio $\frac{p_{oe}}{p_e}$ is obtained.
- 2) The total pressure ratio across the shock is then calculated using:

$$\frac{p_{o2}}{p_{o1}} = \frac{p_{oe}}{p_{o1}} = \frac{p_{oe}}{p_e} \cdot \frac{p_e}{p_{o1}} \quad (4)$$

where $\frac{p_e}{p_{o1}}$ is the known nozzle pressure ratio.

- 3) Using the computed $\frac{p_{o2}}{p_{o1}}$, the upstream Mach number M_1 can be obtained using the following Eq. [2, Equation 6.28]:

$$\frac{p_{o2}}{p_{o1}} = \left(\frac{[(\gamma+1)/2] M_1^2}{1 + [(\gamma-1)/2] M_1^2} \right)^{\gamma/(\gamma-1)} \left[\frac{2\gamma}{\gamma+1} M_1^2 - \frac{\gamma-1}{\gamma+1} \right]^{1/(1-\gamma)} \quad (5)$$

- 4) The corresponding area ratio is then obtained using:

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} \quad (6)$$

- 5) Since $\frac{A_1}{A_1^*} = \frac{A_2}{A_2^*}$, the shock location is thereby established. For all upstream stations before this point the Mach number is computed using Eq. 6. The static thermodynamic properties are determined using Eq. 8 and Eq. 9

Once the location of the normal shock and the upstream Mach number M_1 are known, the downstream Mach number M_2 is computed using the normal shock relation:

$$M_2^2 = \frac{(\gamma-1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma-1)} \quad (7)$$

For all downstream stations, the local Mach number is evaluated using the area-Mach number relation (Eq. (6)), where the corresponding local area is substituted in place of A , and the new critical area A_2^* is used. Once the Mach number has been determined, the static thermodynamic properties are computed using the following isentropic relations:

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}} \quad (8)$$

$$\frac{T_0}{T} = \left(1 + \frac{\gamma-1}{2} M^2 \right) \quad (9)$$

By this approach, the full thermodynamic state along the nozzle, both before and after the shock, is determined.

Numerical Solving

Equations such as Eq. (5) and Eq. (6) are transcendental in nature and do not admit closed-form analytical solutions for the Mach number. To determine the upstream Mach number M_1 , numerical root-finding techniques are employed. In this work, the `fsolve` function from the `scipy.optimize` module in Python is used to obtain the solution.

B. Flowchart

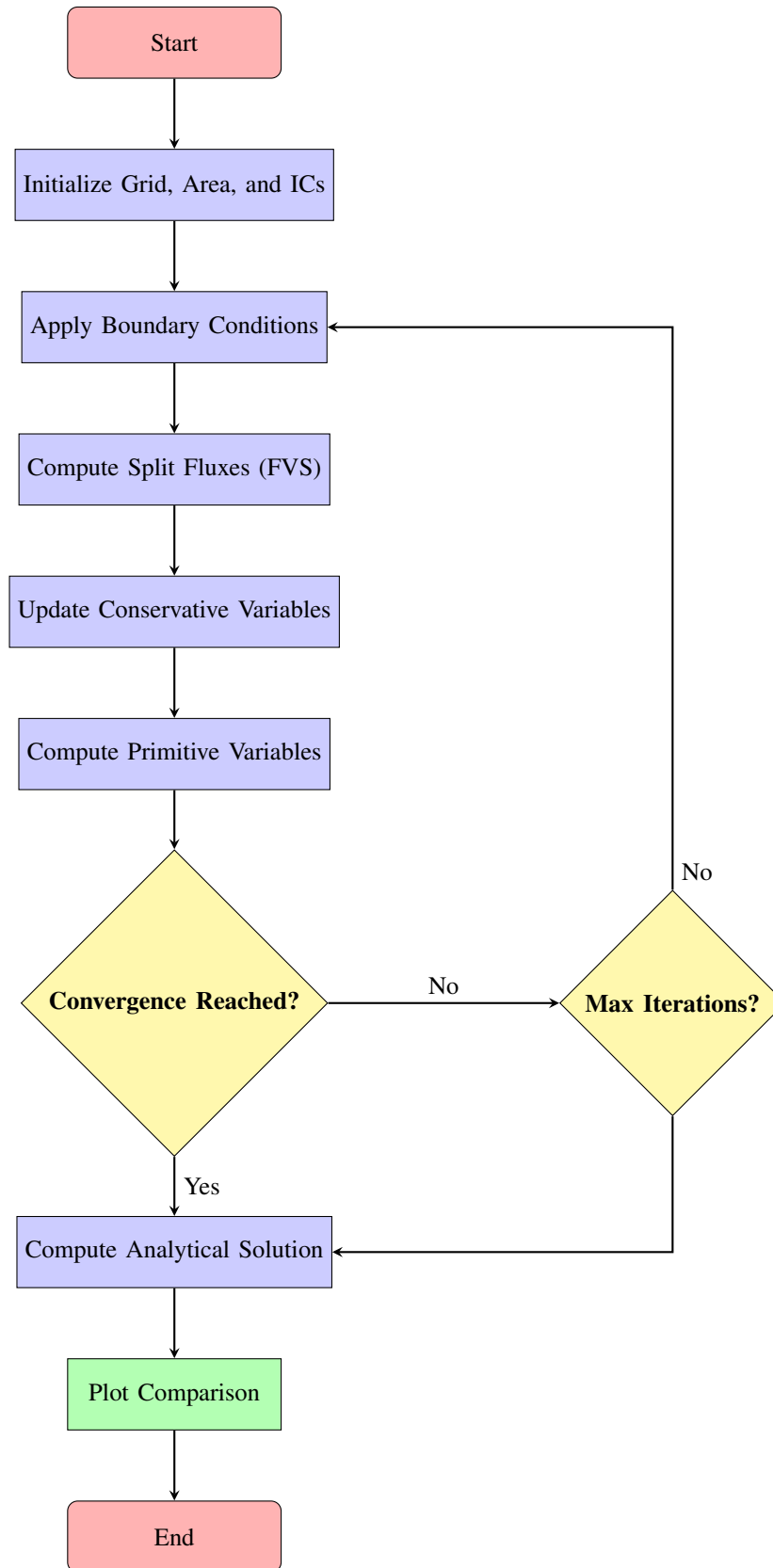


Fig. 3: Flowchart for Compressible Nozzle Flow Solver

II. RESULTS AND DISCUSSION

Convergence:

- Simulation converged in **21,231 iterations**
- Final error: 9.9995×10^{-4}
- Time taken: **10.98 seconds**

Numerical Solution:

- Exit Mach number: **0.34**
- Mach number before the shock: **2.16**
- Mach number after the shock: **0.53**
- Shock location: **1.72 m**

Analytical Solution:

- Exit Mach number: **0.33**
- Mach number just before the shock: **2.20**
- Mach number just after the shock: **0.55**
- Shock location: **1.72 m**

The numerical results show excellent agreement with the analytical solution, particularly in capturing the shock location and strength. Minor discrepancies in Mach numbers are expected due to discretization errors.

A. P/P_0 versus x

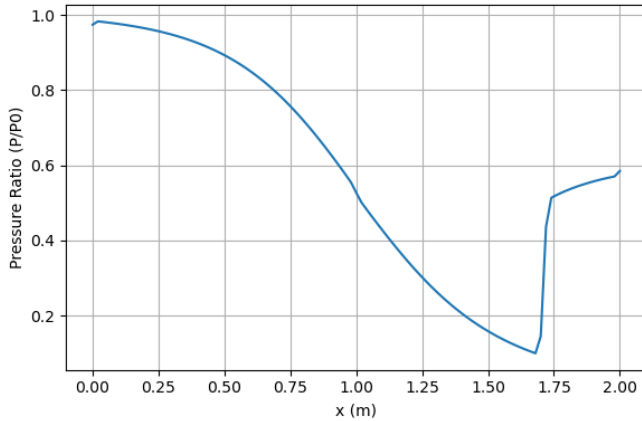


Fig. 4: Variation of pressure ratio along the nozzle axis

Figure 4 illustrates the variation of the pressure ratio P/P_0 along the nozzle axis. Initially, the flow is subsonic and accelerates through the converging section of the nozzle. As expected, the pressure ratio decreases smoothly through the throat due to increasing velocity.

In the diverging section, the flow continues to expand, resulting in a further drop in pressure until a sudden rise is observed. This sharp discontinuity is indicative of a normal shock located within the diverging section. Upstream of the shock, the flow is supersonic; downstream, it becomes subsonic, explaining the abrupt increase in pressure ratio.

Following the shock, the flow remains subsonic and the pressure gradually recovers. The overall profile is consistent with the behavior of a converging-diverging nozzle under overexpanded conditions.

B. Mach Number versus x

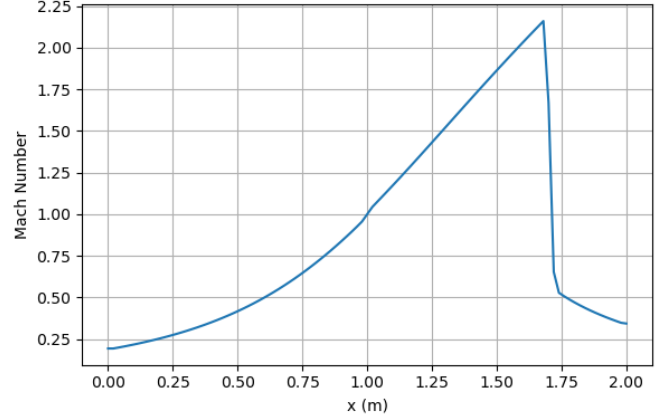


Fig. 5: Variation of Mach number along the nozzle axis

Figure 5 shows the variation of the Mach number along the length of the nozzle. Initially, the flow is subsonic, and the Mach number gradually increases through the converging section. At the throat (at $x = 1.0$ m), the flow reaches Mach 1, marking the transition to sonic conditions.

In the diverging section, the flow accelerates further into the supersonic regime, as expected for a properly expanded nozzle. However, a sharp drop in Mach number is observed around $x \approx 1.7$ m, which signifies the presence of a normal shock.

Across the shock, the supersonic flow abruptly transitions back to subsonic, consistent with the physics of shock waves in compressible flow. After the shock, the Mach number continues to decrease smoothly due to subsonic deceleration in the remaining diverging section.

This behavior confirms the presence of a normal shock within the nozzle, which is typical in overexpanded operating conditions.

C. Comparison of Numerical and Analytical Results

Figure 6 presents a comparative analysis of the numerical results against analytical and idealized solutions for Mach number, pressure ratio (P/P_0), and temperature ratio (T/T_0) distributions along the length of the converging-diverging nozzle.

In the top subplot, the Mach number distribution reveals excellent agreement between the numerical and analytical solutions in both subsonic and supersonic regions. The flow accelerates smoothly from subsonic to sonic conditions at the throat and continues to supersonic speeds in the diverging section. The numerical solution captures the normal shock at approximately $x = 1.72$ m, where a sudden drop in Mach number occurs. The analytical curve also reflects this transition. The idealized curves (supersonic and subsonic) serve as bounding solutions, showing the two possible paths the flow could follow in the absence of shock.

The middle subplot shows the pressure ratio (P/P_0) distribution. Upstream of the shock, all models follow a similar

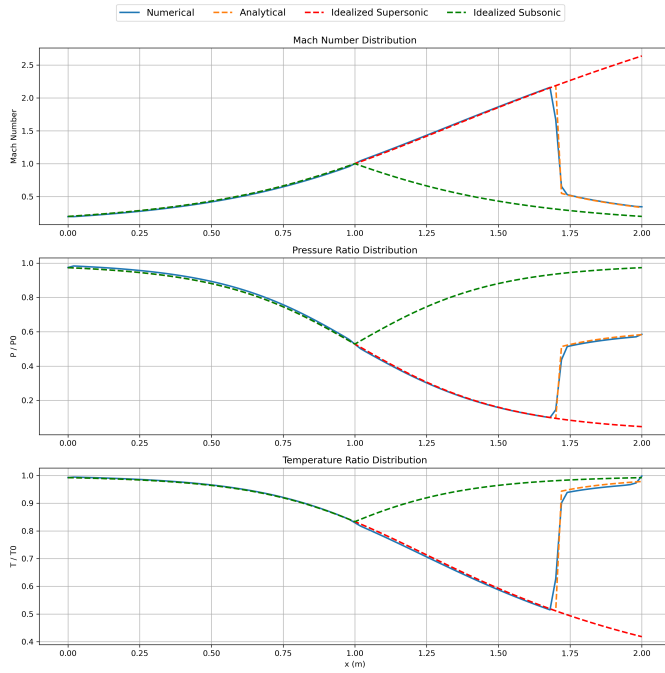


Fig. 6: Comparison of numerical results with analytical and idealized solutions along the nozzle axis

decreasing trend, consistent with isentropic expansion. Across the shock, the pressure rises sharply in the numerical and analytical results, validating the presence of a shock. The idealized supersonic and subsonic solutions, however, continue their smooth paths without accounting for this shock behavior.

In the bottom subplot, the temperature ratio (T/T_0) exhibits similar behavior. A steady decrease is observed in the supersonic expansion region, followed by a rapid increase across the shock due to compression and flow deceleration. Again, the numerical solution closely follows the analytical prediction, with deviations only near the discontinuity where the shock is numerically smeared.

Overall, the numerical solution accurately captures the flow physics, including isentropic expansion, shock formation,

REFERENCES

- [1] J. D. Anderson, *Modern Compressible Flow With Historical Perspective*. McGraw Hill, 2020, ch. 5, pp. 216–217 (cit. on p. 2).
- [2] R. T. Zucker and O. Biblarz, *Fundamentals of Gas Dynamics*, 3rd. John Wiley & Sons, 2020, ch. 6, p. 169 (cit. on p. 2).